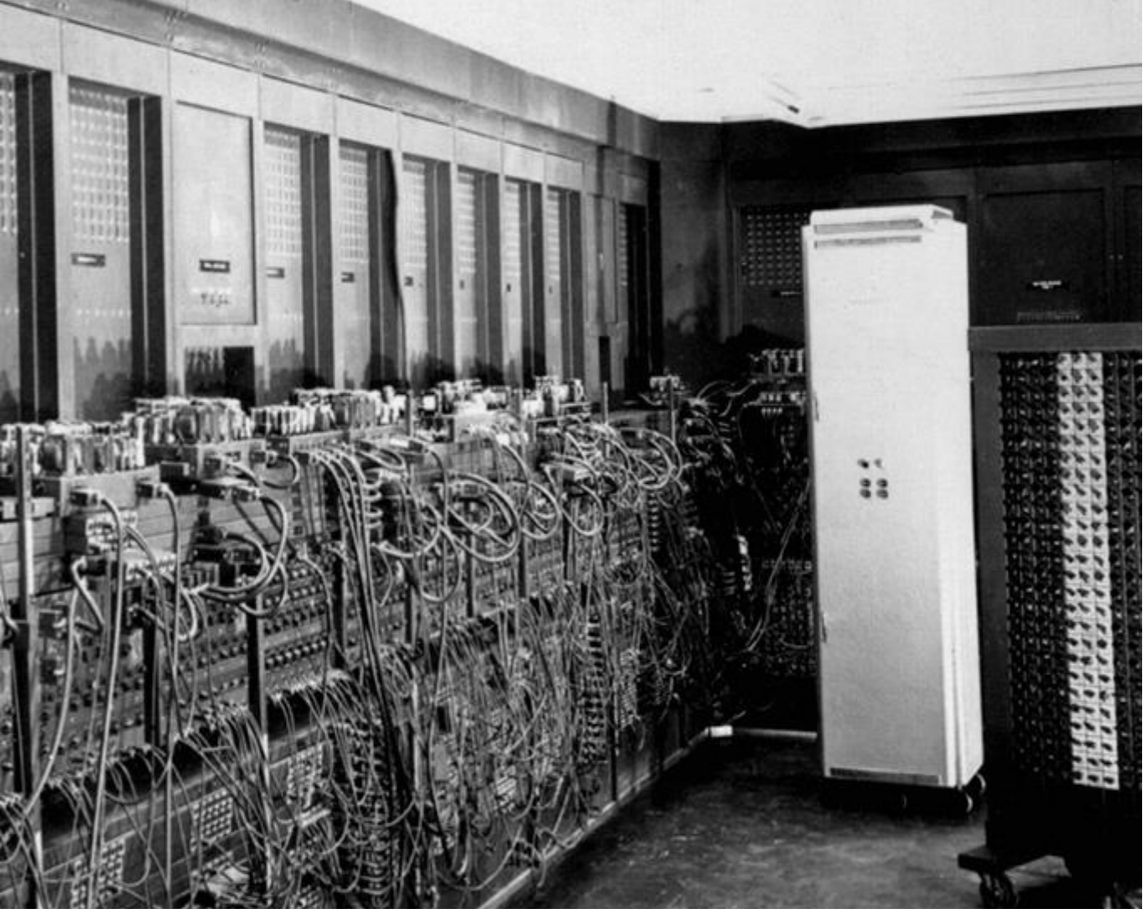


EIU

Week 1: Lesson 1

Foundations of Cloud Computing and EC2



amazon EC2

Rohit Kumar Kasera
Lecturer

Agenda

01

History of computing

The first ENIAC general-purpose electronic computer and its revolutionary impact

02

Evolution of Computer Hardware

From vacuum tubes to microprocessors and the miniaturization revolution

03

The Rise of Data Centers

Centralized computing power and the challenges it presented

04

Introduction to virtual machines & cloud computing

The paradigm shift that transformed the computing landscape forever

05

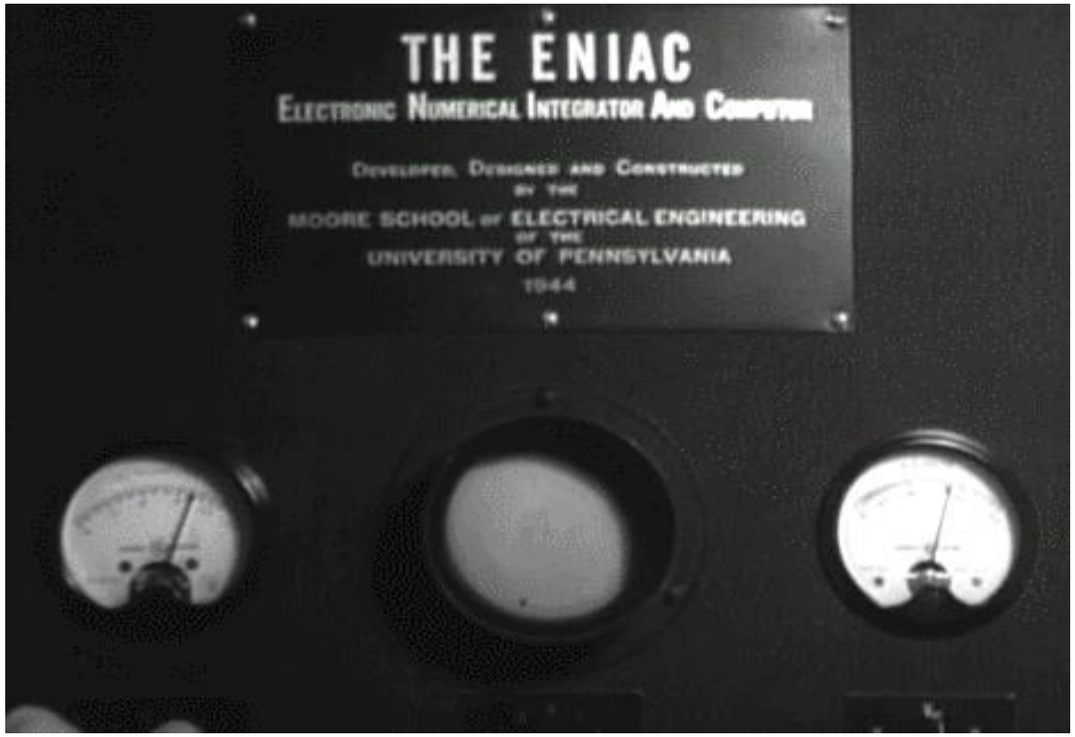
Amazon Global Cloud Infrastructure

A service that provides resizable virtual servers in the cloud.

Today, we'll explore how computing evolved from massive, specialized machines to the flexible, virtualized resources we use daily. We'll examine key technological breakthroughs and their far-reaching implications.

History of computing

A journey through the remarkable transformation of computing technology that has reshaped our world, from room-sized military calculators to powerful machines in our pockets and virtual servers across the globe.



The Birth of Modern Computing

ENIAC: The Electronic Numerical Integrator and Computer

Why Was ENIAC Built?

During World War II, the U.S. Army faced a critical challenge: calculating artillery firing tables for battlefield accuracy. These complex calculations required:

- Thousands of mathematical operations for a single trajectory
- Calculations adjusted for different weapons, terrain, and weather conditions
- Speed and precision that human "computers" couldn't consistently deliver

  - The military recognized that electronic computation could solve this problem, leading to the commissioning of ENIAC as part of the war effort in 1943.
  - Starting point of electronic computing.



Human "computers" – typically women with mathematics backgrounds – manually calculated firing tables before ENIAC

ENIAC: A Technical Marvel

18,000

Vacuum Tubes

The electronic switches that made digital computing possible

70,000

Resistors

Components that regulated electrical current flow

10,000

Capacitors

Components that stored electrical charge

1,800

Square Feet

Physical space required to house the massive machine

The ENIAC cost between approximately $400,000 and $487,000 to build, a figure that was equivalent to roughly $5.9 million to $6.9 million in today dollars. ENIAC consumed about 150 kilowatts of power and weighed nearly 30 tons. Despite its size, it could only perform about 5,000 additions per second – millions of times slower than today's smartphones. Still, it was approximately 1,000 times faster than the electromechanical machines that preceded it.

Programming ENIAC: A Physical Challenge



Programming ENIAC was nothing like modern coding:

- No keyboards, screens, or programming languages existed
- Each new problem required physically rewiring the machine
- Programmers manually set dials and plugged cables into panels
- A single program setup could take days or weeks
- Debugging meant checking thousands of physical connections

This physical programming process highlighted the need for stored-program computers, which would soon revolutionize computing.

The Evolution of Computer Hardware

From Room-Sized Machines to Pocket-Sized Devices

EIU

The Miniaturization Revolution



1

1940s-50s: Vacuum Tube Era

Room-sized computers like ENIAC and UNIVAC. Expensive, unreliable, and required constant maintenance.

2

1950s-60s: Transistor Era

Smaller, more reliable solid-state transistors replaced vacuum tubes, reducing size and power consumption dramatically.

3

1960s-70s: Integrated Circuit Era

Multiple transistors combined on silicon chips, enabling minicomputers like the PDP-11 that could fit in a small room.

4

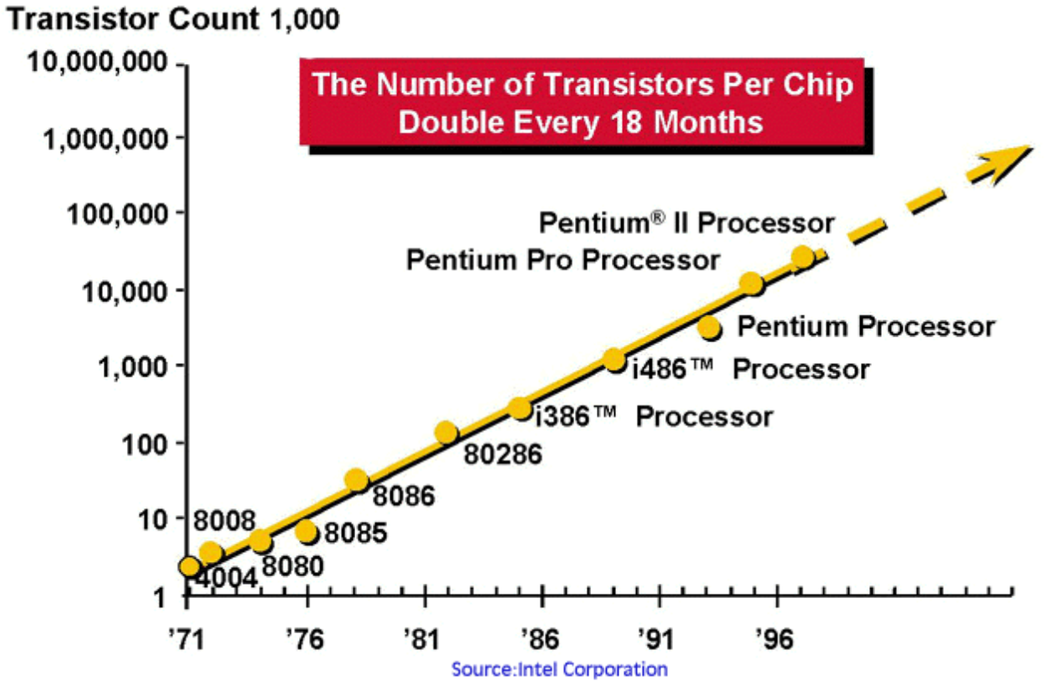
1970s-Present: Microprocessor Era

Entire CPUs on a single chip, leading to personal computers, laptops, smartphones, and wearables.

Moore's Law: The Engine of Computing Progress

> "The number of transistors on an integrated circuit doubles approximately every two years."
>
> — Gordon Moore, Intel co-founder, 1965

- This observation, which became known as Moore's Law, has held remarkably true for over 50 years, driving exponential growth in computing power while costs have steadily decreased.
- Every two years, computers have typically become twice as powerful at the same price point.



The exponential growth in transistor density has enabled the computing revolution we've experienced

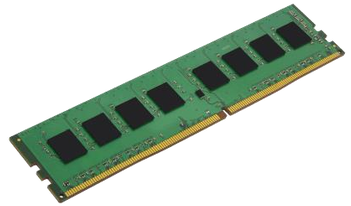
Core Hardware Components

Central Processing Unit (CPU)



The "brain" that executes instructions and performs calculations. Modern CPUs contain billions of transistors on a chip smaller than a postage stamp.

Random Access Memory (RAM)



Fast temporary storage for active programs and data. Unlike storage, RAM is volatile and loses its contents when power is off.

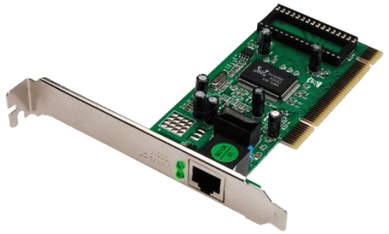
Storage (HDD/SSD)





Permanent data storage using either mechanical hard disk drives (HDDs) or faster solid-state drives (SSDs) with no moving parts.

Network Interface Card (NIC)



Enables connectivity to networks and the internet, transforming computers from isolated machines to connected devices.

These components, working together with software (operating systems and applications), form the foundation of modern computing systems, from smartphones to supercomputers.

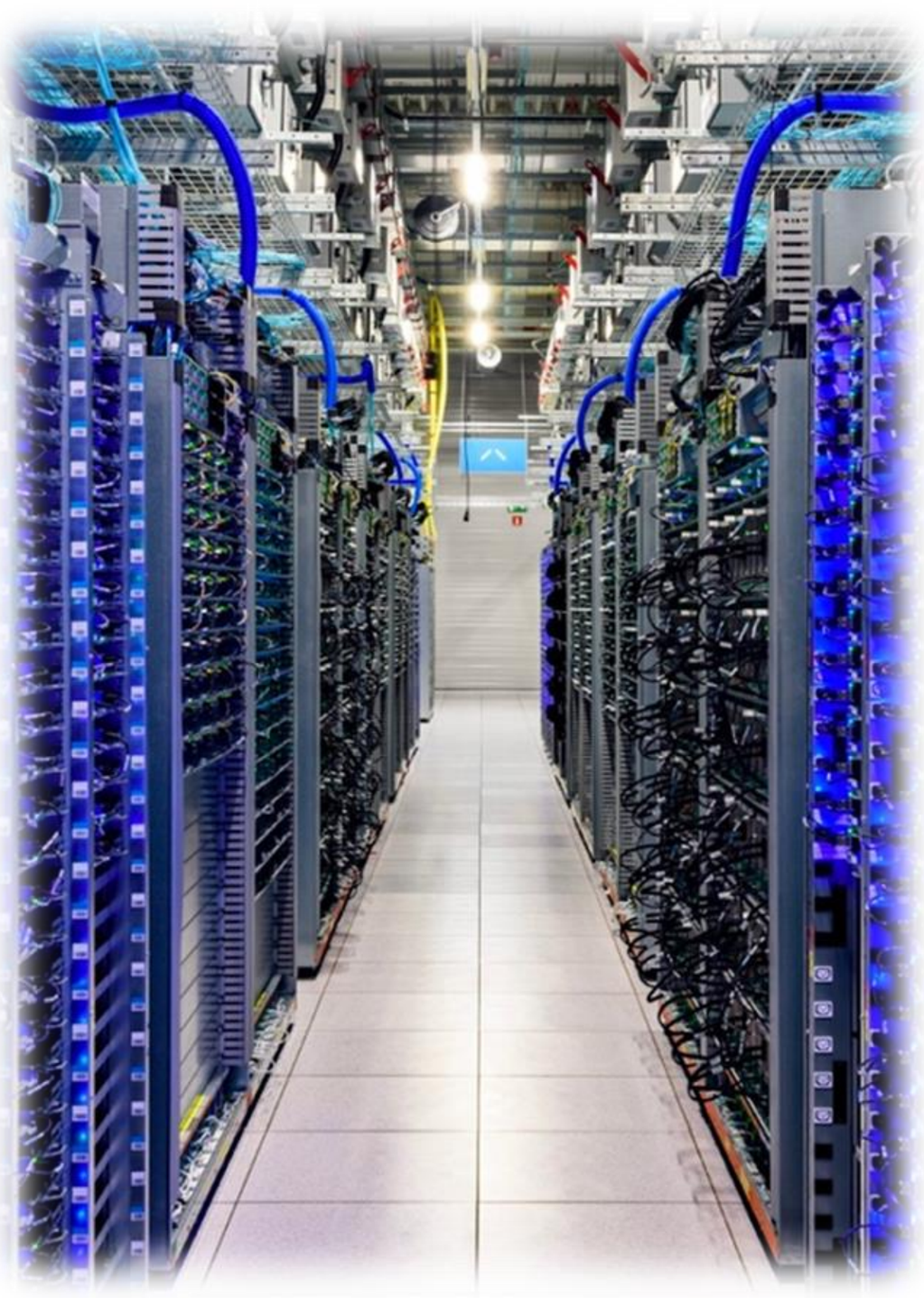
The Democratization of Computing

The rapid evolution of hardware led to unprecedented democratization of computing power:

- **1940s-50s:** Computers cost millions, used only by militaries and large institutions
- **1960s-70s:** Minicomputers for thousands of dollars enabled business adoption
- **1980s:** Personal computers brought computing to homes and small businesses
- **1990s-2000s:** Internet-connected devices expanded computing's reach
- **2010s-Present:** Mobile and cloud computing put enormous computational power in everyone's hands



Today's smartphone has more computing power than all of NASA had during the Apollo moon missions.



The Rise of the Data Center

Centralizing Computing Power

Why Data Centers Emerged

As computing became essential to business operations, organizations faced new challenges that individual computers couldn't solve:



Growing Demand

Business applications required more processing power than single machines could provide

Network Needs

Organizations needed central locations to manage networked resources and data

Data Growth

Exploding volumes of digital information required specialized storage infrastructure

The solution was the data center: a specialized facility housing computing, storage, and networking equipment in a controlled environment with dedicated power, cooling, and security systems.

Evolution of Data Centers

1

1970s-80s: Early Corporate Data Centers

Centralized mainframe computers from companies like IBM, used primarily for business operations, financial processing, and record keeping.

2

1990s: Internet Service Providers

Companies like AOL and telecom providers built large server rooms to support growing internet services and web hosting.

3

2000s: Enterprise Data Centers

Organizations built dedicated facilities with raised floors, advanced cooling, and redundant power to support mission-critical applications.

4

2010s-Present: Hyperscale Data Centers

Tech giants like Google, Microsoft, and Amazon operate massive facilities with hundreds of thousands of servers, often in remote locations near renewable energy sources.

Inside a Modern Data Center



- **Server Racks:** Standardized frames housing computing equipment
- **Network Infrastructure:** Switches, routers, and cabling connecting servers
- **Storage Systems:** Specialized equipment for data storage and backup
- **Power Systems:** Uninterruptible power supplies and backup generators
- **Cooling Systems:** Precision air conditioning to prevent overheating
- **Security:** Physical access controls, fire suppression, and monitoring

Modern hyperscale data centers can contain hundreds of thousands of servers and consume as much electricity as a small city.

The Economics of Data Centers

$600M	$1.2B	3-5	24/7
Minimum Cost	Typical Cost	Years	Staffing
Starting cost for a Google-scale data center	Average investment for a hyperscale facility	Typical hardware refresh cycle requiring reinvestment	Round-the-clock expert staffing requirements

Traditional data centers represented massive capital investments with ongoing operational costs. Companies needed to predict their computing needs years in advance, often leading to overprovisioning (wasting resources) or under provisioning (limiting growth).

Challenges of Traditional Data Centers

Capital Expense Burden

Huge upfront investments in buildings, power infrastructure, cooling systems, and hardware, often depreciating over 15-20 years

Slow Scalability

Expanding capacity required lengthy procurement processes, construction, and installation – often taking months or years

Operational Inefficiency

Average server utilization in traditional data centers was only 12-18%, meaning most computing capacity sat idle most of the time

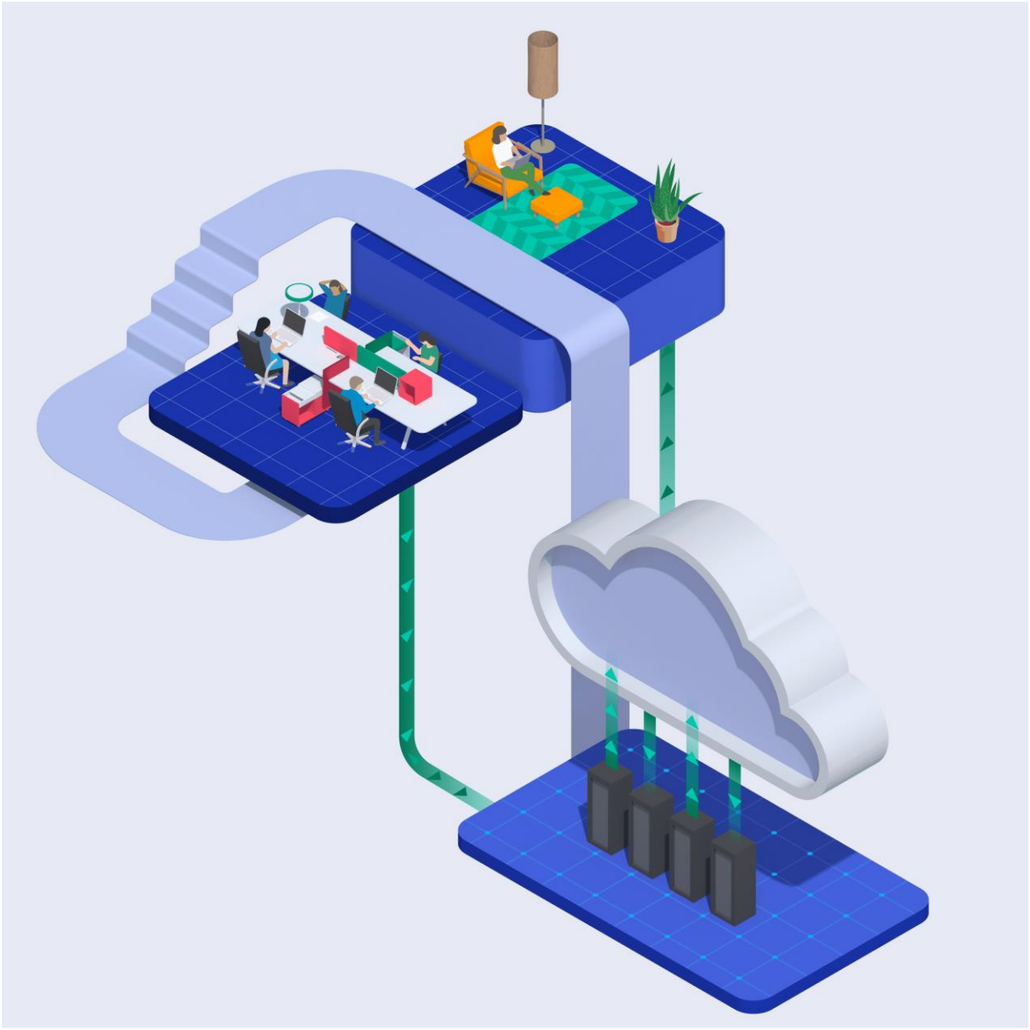
Environmental Impact

High energy consumption and carbon footprint, with significant cooling requirements and power inefficiencies

These limitations highlighted the need for a more flexible, efficient approach to computing infrastructure.

Virtual Machines & Cloud Computing

The Paradigm Shift

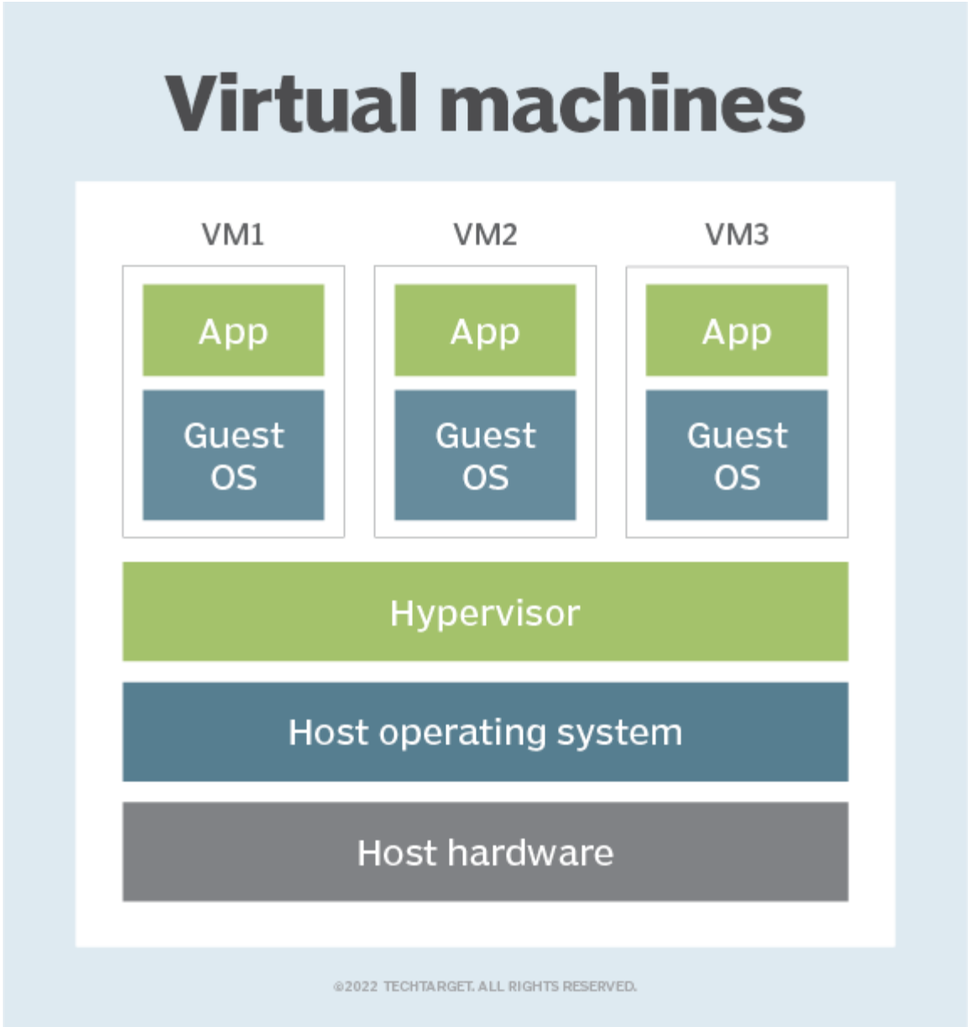


What Is a Virtual Machine?

A virtual machine (VM) is a software-based emulation of a physical computer that:

- Runs its own operating system and applications
- Behaves exactly like a physical computer
- Is completely isolated from other VMs on the same hardware
- Can be created, copied, moved, or deleted with simple commands

VMs transformed computing by breaking the one-to-one relationship between physical hardware and operating systems, allowing multiple "computers" to run on a single physical machine.



How Virtual Machines Work

01

Hypervisor Installation

A specialized software layer called a hypervisor (like VMware ESXi, Microsoft Hyper-V, or KVM) is installed on the physical server

02

Resource Allocation

The hypervisor divides the physical server's CPU, memory, storage, and network resources into multiple virtual "slices"

03

VM Creation

Virtual machines are created as isolated containers, each with its own virtual hardware

04

Operating System Installation

Each VM runs its own complete operating system, which sees only its allocated virtual resources

The hypervisor manages all VM operations and ensures they remain isolated from each other while efficiently sharing the underlying physical hardware.

Benefits of Virtual Machines



Resource Efficiency

Multiple VMs share a single physical server, increasing utilization from 15% to 80%+ and reducing hardware costs



Isolation & Security

Applications in separate VMs can't interfere with each other, improving reliability and security



Portability

VMs can be moved between physical servers without reconfiguration, enabling maintenance without downtime



Agility

New VMs can be deployed in minutes rather than the weeks or months required for physical servers

Virtual machines fundamentally changed the relationship between software and hardware, enabling the flexibility that would eventually make cloud computing possible.

What Is Cloud Computing?

Cloud computing takes the flexibility introduced by VMs a step further, transforming computing infrastructure into a utility. It is the on-demand delivery of computing power, database storage, applications, and other IT resources via the internet with pay-as-you-go pricing.

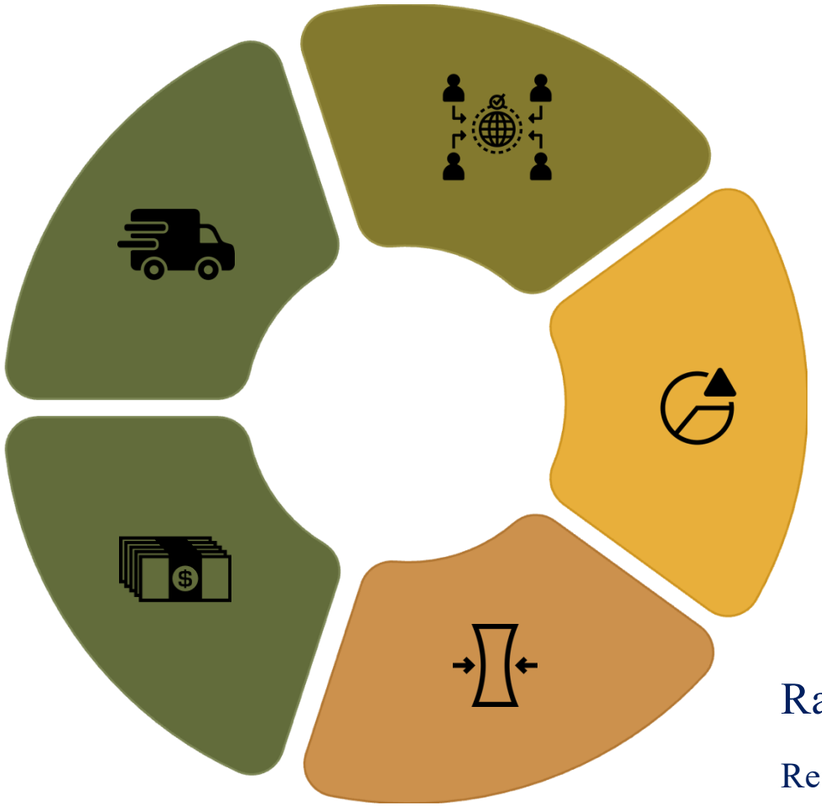
On-Demand Self-Service

Resources available instantly without human intervention

- Compute Storage
- Databases
- Networking
- AI/ML Security

Measured Service

Pay only for resources consumed



Broad Network Access

Services accessible from any internet-connected device

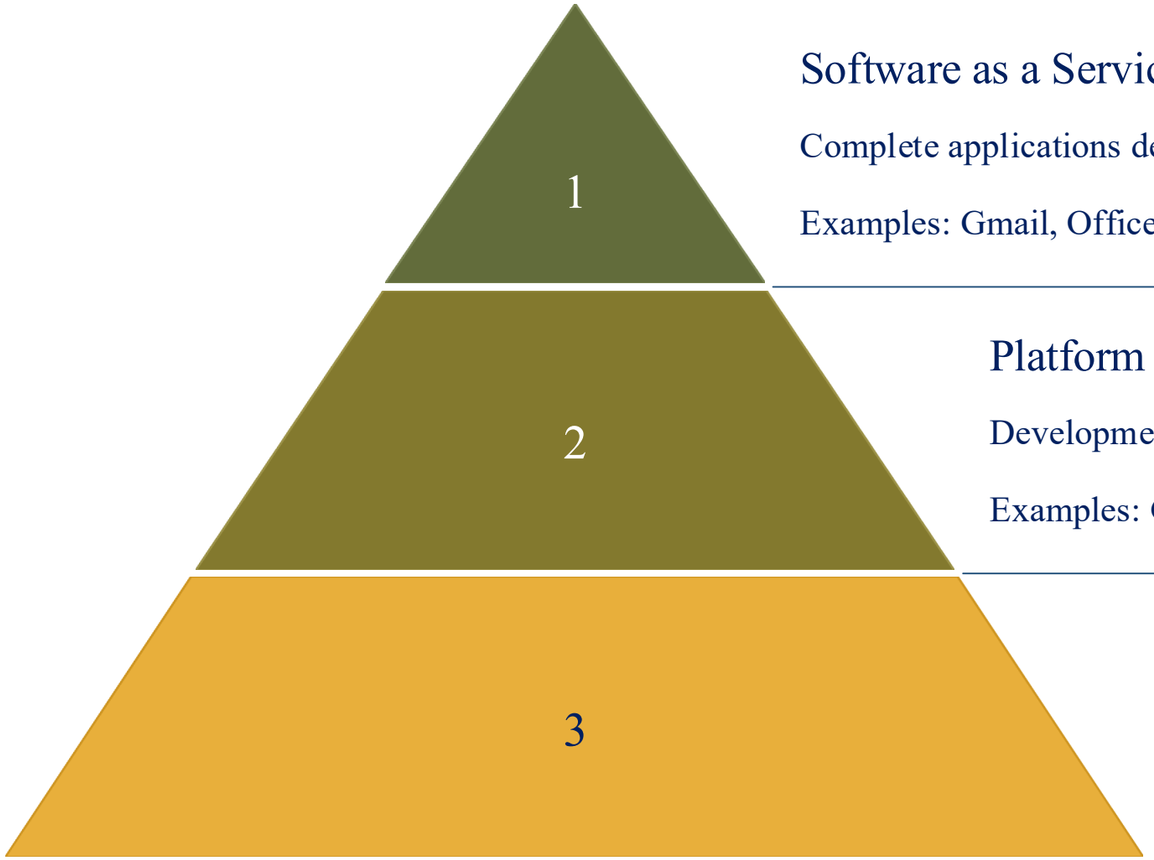
Resource Pooling

Shared infrastructure serving multiple customers efficiently

Rapid Elasticity

Resources scale up or down automatically with demand

Cloud Service Models



Software as a Service (SaaS)

Complete applications delivered over the web

Examples: Gmail, Office 365, Salesforce

Platform as a Service (PaaS)

Development and deployment environment for building applications

Examples: Google App Engine, Heroku, Azure App Service

Infrastructure as a Service (IaaS)

Basic computing resources: servers, storage, networking

Examples: AWS EC2, Azure VMs, Google Compute Engine

Each service model represents a different level of abstraction, with SaaS offering the most managed experience and IaaS providing the most control and responsibility for the customer.

The Pioneers of Cloud Computing

1999: Salesforce

Pioneered Software as a Service (SaaS) with its cloud-based CRM system, proving that business applications could be delivered over the internet

1

2006: Amazon Web Services

Launched S3 (storage) and EC2 (compute), revolutionizing infrastructure by making virtual servers available on-demand with pay-per-use pricing

2

2008: Google App Engine

Introduced Platform as a Service (PaaS), allowing developers to build and deploy applications without managing servers

3

2010: Microsoft Azure

Brought cloud services to enterprise environments, bridging traditional Microsoft software with cloud capabilities

4

These pioneers transformed computing from a product you buy to a service you consume, fundamentally changing how technology is built, deployed, and managed.

Real-world examples

- Netflix
- Uber
- Spotify
- E-commerce platforms
- AI chatbots

Benefits

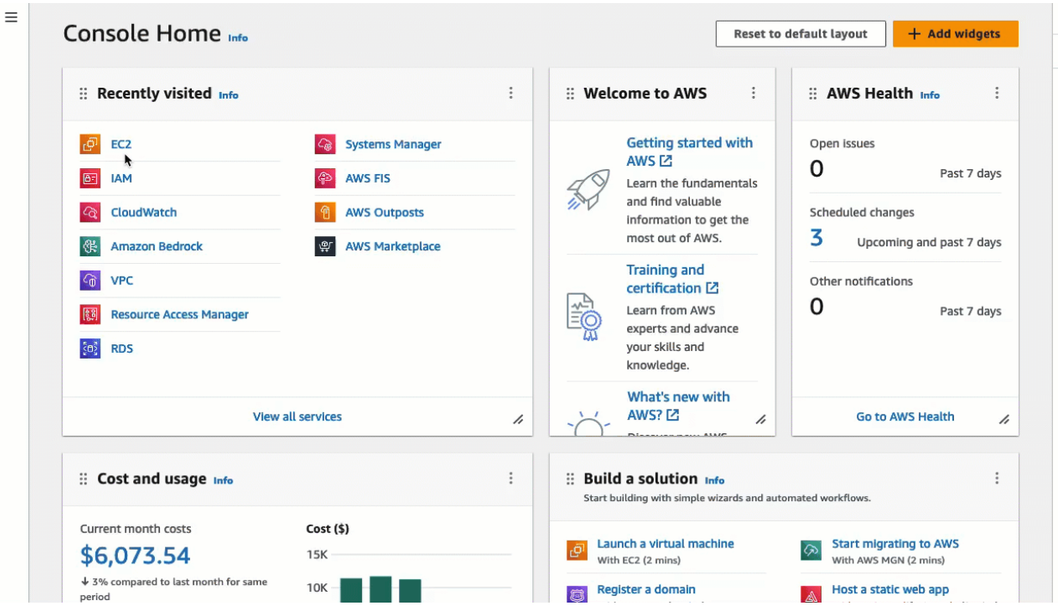
- Lower cost
- No hardware purchase
- Global deployment
- High availability
- Auto scaling
- Security services
- Backup and disaster recovery
- Faster innovation

Amazon Global Cloud Infrastructure

- AWS currently spans 39 geographic Regions and 123 Availability Zones globally.
- When Amazon launched Elastic Compute Cloud (EC2) in 2006, it fundamentally changed computing:

- **Virtual Servers On-Demand:** Spin up new servers in minutes, not months
- **No Capital Investment:** Pay only for what you use, when you use it
- **Elastic Scaling:** Automatically add or remove capacity based on demand
- **Global Reach:** Deploy workloads in data centers around the world
- **Diverse Instance Types:** Specialized configurations for different workloads

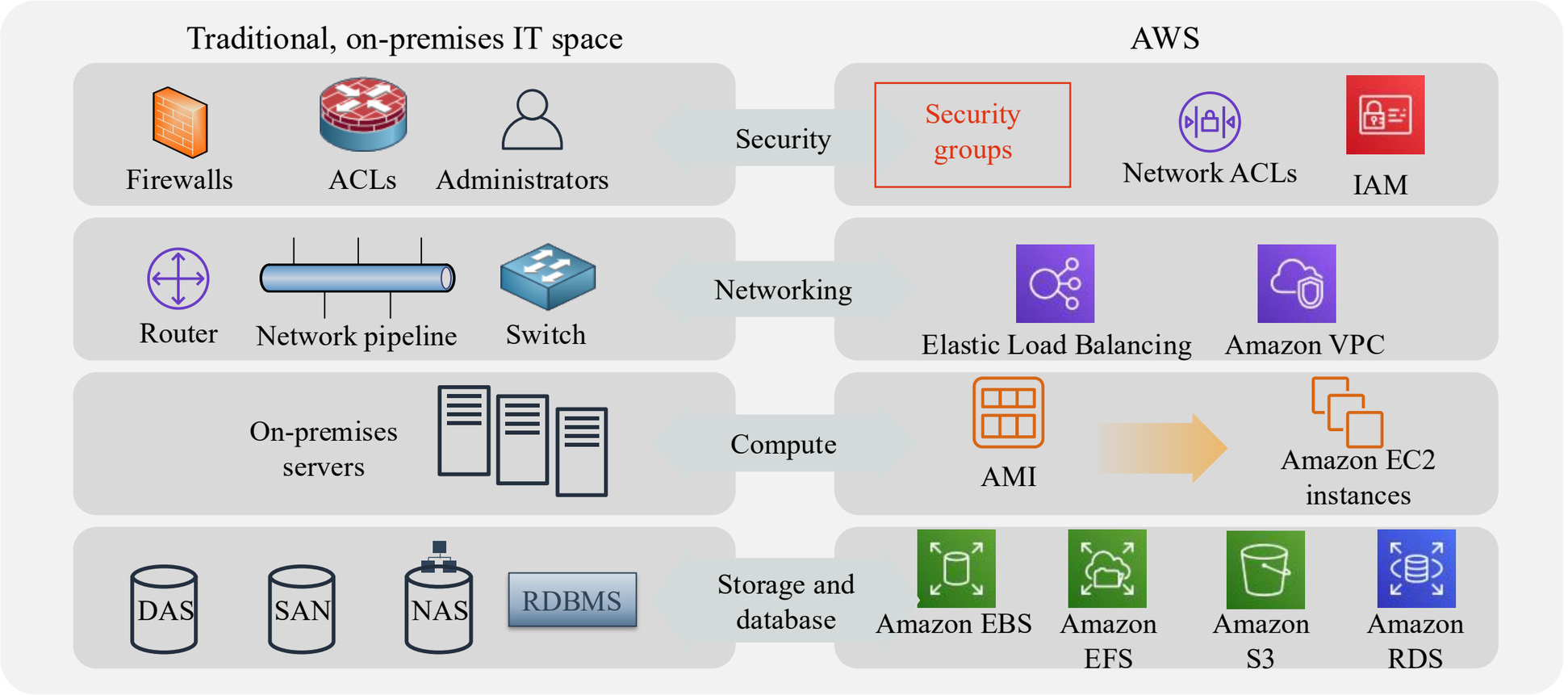
EC2 enabled startups to compete with enterprises by giving them access to the same infrastructure without massive upfront investments.



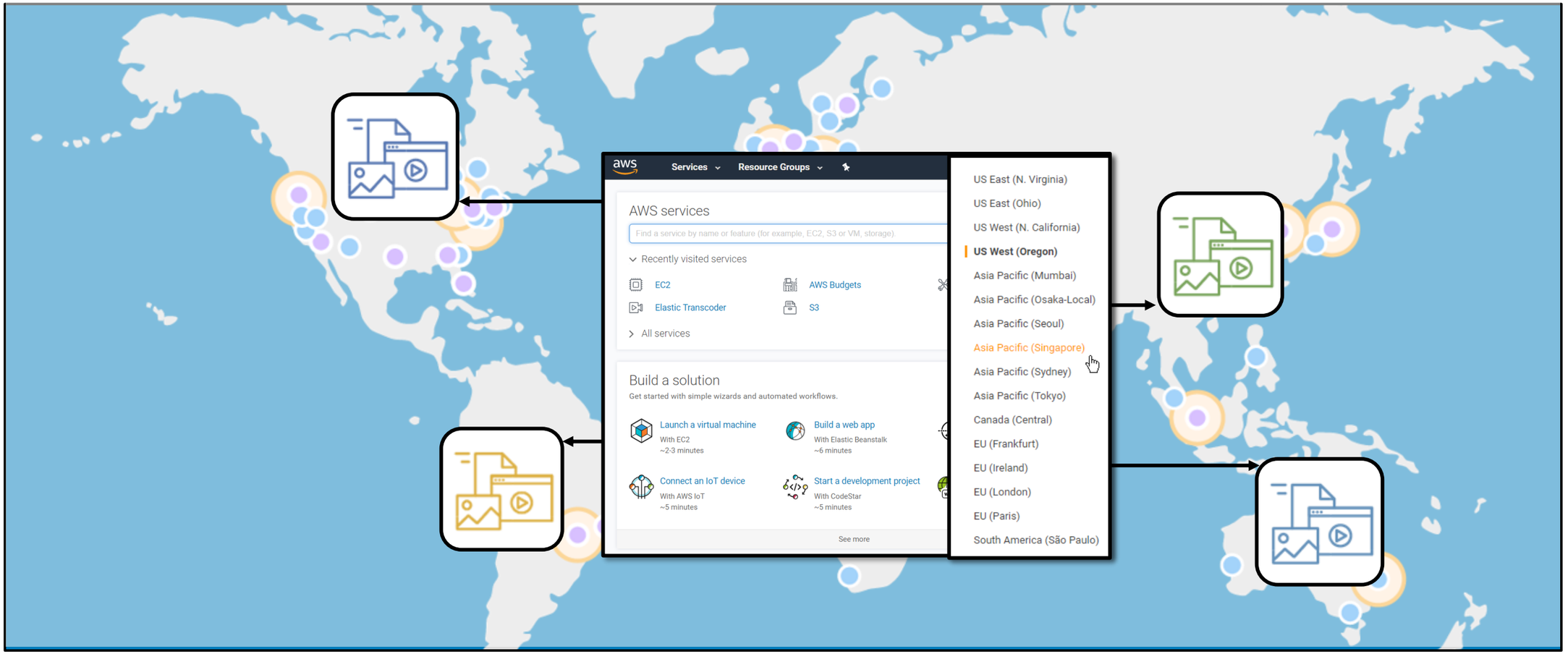
Use cases

- ✓ Web hosting
- ✓ Databases
- ✓ AI workloads
- ✓ Development/testing
- ✓ Big data processing
- ✓ Game servers

Similarities between AWS and traditional IT



Go global in minutes



From ENIAC to EC2: The Transformation

ENIAC → Mainframes → Servers → Virtual Machines → Cloud

Physical to Virtual

Computing evolved from requiring specific hardware for each task to software-defined resources that can run anywhere

Local to Distributed

Processing moved from single locations to globally distributed systems working in concert

Static to Dynamic

Resources transformed from fixed, unchanging configurations to systems that adapt in real-time to changing demands

Expensive to Cost-Efficient

Computing shifted from massive capital investments to pay-as-you-go operational expenses aligned with actual usage

- In less than a century, computing evolved from room-sized calculators for military ballistics to virtual servers that anyone can deploy globally with a few clicks, democratizing access to computational power.

Categories of AWS services


Analytics


Application Integration


AR and VR


Blockchain


Business Applications


Compute


Cost Management


Customer Engagement


Database


Developer Tools


End User Computing


Game Tech


Internet of Things


Machine Learning


Management and Governance


Media Services


Migration and Transfer


Mobile


Networking and Content Delivery


Robotics


Satellite


Security, Identity, and Compliance


Storage

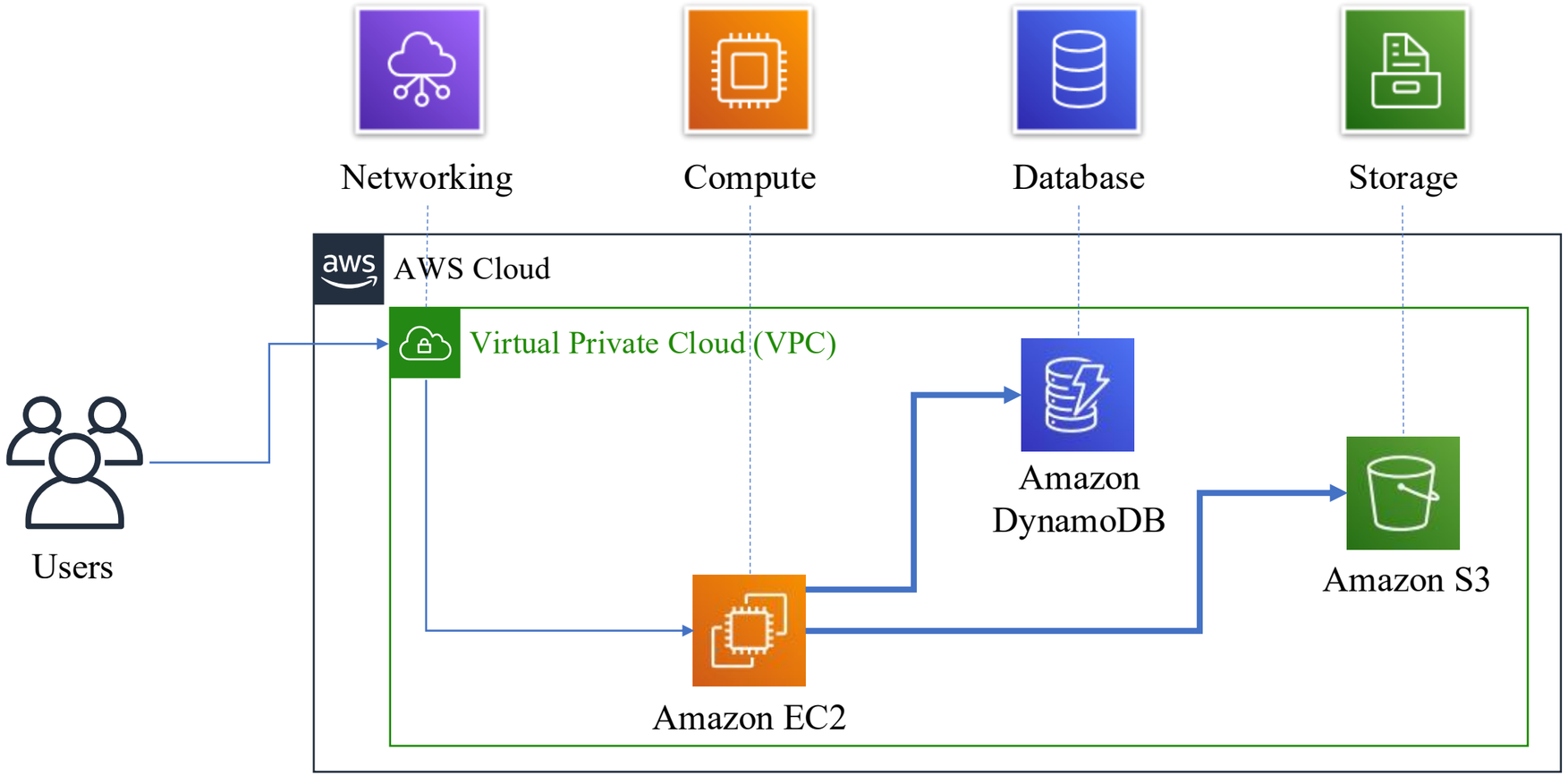
Important EC2 Features

- Multiple instance types
- Different OS options
- Auto scaling
- Load balancing
- Security groups
- Elastic IP
- AMIs
- EBS storage
- Spot and Reserved instances

Common instance families

- General purpose → t3, t4g
- Compute optimized → c7g
- Memory optimized → r7g
- GPU → p-series

Simple solution example



The Future of Computing

As we look forward, several trends are shaping the next evolution of computing:

- **Edge Computing:** Processing moving closer to data sources
- **Serverless Computing:** Focus on code, not infrastructure
- **Quantum Computing:** New paradigms for solving complex problems
- **AI & Machine Learning:** Autonomous systems and intelligence
- **Sustainable Computing:** Reducing environmental impact

The pace of innovation continues to accelerate, with each breakthrough building on the foundation laid by ENIAC and its successors.

